

Coding Standards — Requirements

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Applies To: All software developed and maintained by the team

1. General Requirements

- Code **MUST** be clear, readable, and maintainable.
- Code **MUST** follow the established conventions of the project.
- Developers **MUST** avoid unnecessary complexity and duplication.
- Unused code, variables, imports, and dependencies **MUST** be removed.
- Code **SHOULD** follow the principle of single responsibility.

2. Naming and Formatting

- Variables, functions, classes, and methods **MUST** have clear and descriptive names.
- Abbreviations **SHOULD** be avoided unless commonly understood.
- Code **MUST** follow the project's formatting and indentation standards.
- Automated formatting tools **SHOULD** be used when available.
- Unnecessary formatting changes **SHOULD NOT** be included with unrelated code changes.

3. Comments and Documentation

- Comments **MUST** explain non-obvious behavior or design decisions.
- Comments **MUST NOT** simply restate what the code does.
- Public APIs and complex functionality **MUST** have appropriate documentation.
- Outdated comments **MUST** be updated or removed.

4. Error Handling and Logging

- Errors **MUST** be handled explicitly and appropriately.
- Errors **MUST NOT** be silently ignored.
- Error messages **MUST NOT** expose sensitive information.

- Applications **MUST** use the approved logging and error-handling mechanisms.
- Logs **MUST NOT** contain passwords, tokens, secrets, or other sensitive information.
- Logs **SHOULD** contain enough information to troubleshoot problems.

5. Security

- Secrets, credentials, and API keys **MUST NOT** be stored in source code.
- Untrusted input **MUST** be appropriately validated.
- Authentication and authorization **MUST** follow approved security requirements.
- Security controls **MUST NOT** be disabled without authorization.
- Dependencies **MUST** be evaluated for known security vulnerabilities.

6. Testing

- New functionality **MUST** include appropriate automated tests.
- Bug fixes **SHOULD** include tests that verify the correction.
- Tests **MUST** be repeatable and deterministic.
- Required tests **MUST** pass before code is merged.
- Important error and boundary conditions **SHOULD** be tested.

7. Version Control

- All production code **MUST** be maintained in the approved version-control system.
- Commits **MUST** represent logical changes and have meaningful messages.
- Secrets and confidential information **MUST NOT** be committed.
- Pull requests **MUST** clearly describe the proposed changes.
- Pull requests **MUST** pass required automated checks before merging.
- Unrelated changes **SHOULD NOT** be included in a pull request.

8. Code Review

- Production code **MUST** undergo the required code-review process.
- Reviewers **MUST** evaluate correctness, maintainability, security, and testing as appropriate.

- Required review issues **MUST** be resolved before approval.
- Code **MUST NOT** be approved by the author when independent review is required.

9. Dependencies and Configuration

- New dependencies **MUST** have an appropriate technical justification.
- Dependencies **MUST** comply with security and licensing requirements.
- Unused dependencies **MUST** be removed.
- Environment-specific configuration **MUST NOT** be hard-coded.
- Secrets **MUST** be managed using approved mechanisms.

10. AI-Assisted Development

- AI-generated code **MUST** be reviewed by a developer before being committed.
- Developers **MUST** understand and verify code they commit.
- AI-assisted code **MUST** meet the same security, testing, formatting, and quality standards as other code.
- Confidential information **MUST NOT** be submitted to AI tools unless explicitly approved.

11. Compliance and Exceptions

- Developers **MUST** comply with the requirements in this document.
- Automated tools **SHOULD** be used to enforce standards where practical.
- Exceptions **MUST** have a documented technical justification and appropriate approval.
- This document **SHOULD** be reviewed periodically and updated as needed.

12. Requirement Definitions

- **MUST / MUST NOT** — Mandatory requirement or prohibition.
- **SHOULD / SHOULD NOT** — Recommended practice that may be deviated from with reasonable justification.
- **MAY** — Optional practice.